

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE)

December 2011

CE 208: Object Oriented Concepts and Visual C++ Programming

Date: 07.12.2011, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Answer following questions: [07]

1. How does Class accomplish data hiding? 01
2. Which operators you can't overload? 01
3. What is Inline Function? 01
4. What is use of destructor? Is it compulsory to use destructor in program? 01
5. Find errors, if any in the following two statements: 01
long float x;
const int array_size;
6. State whether the following statement is True or False, give reason for same 01
"A function can return a value by reference."
7. When we make a virtual function "pure"? Explain briefly. 01

Q - 2 Attempt following : [14]

- (a) What is friend function? Write a VC++ program to demonstrate how friend function can modify the object passed as an argument to friend function. 07
- (b) What is inheritance in VC++? Explain different forms of inheritance with example. 07
Write a syntax of the single inheritance in VC++.

OR

Q - 2 Attempt following : [14]

- (a) Write a VC++ program to create a Class Car that has Model_no, Name, Color and Cost as Data member and Create *void Get_cardetails(void)* and *void Show_cardetails(void)* as member function. 06
- (b) Explain terms with example in VC++: Static binding and Late binding 04
- (c) What is function overloading? What is significance of function overloading in OOP? 04

Q - 3 Attempt following : [14]

- (a) What is static member function? Write a VC++ program to demonstrate the use of Static member function. 05
- (b) How to use constructor in VC++? Explain constructor overloading with example. 05
- (c) Briefly explain Windows program structure. 04

OR

Q - 3 Attempt following : [14]

- (a) Write a program in VC++ to define a *class String* and use overloaded *=* operator to compare two strings. 05
- (b) What MFC? Explain document/view architecture in detail. 05
- (c) List and explain various file operations and file mode options available. 04

SECTION – II

- Q - 4 Attempt following :** [07]
1. What is Abstract class? 01
 2. What is UML? List various diagrams used in UML. 02
 3. What is Composition relationship in Class diagram? Give one example of it. 02
 4. How OOAD is differing from OOP? 02

- Q - 5 Attempt following :** [14]
- (a) Explain concept of call by reference in VC++ with example. 05
 - (b) Explain Copy Constructor in brief. 03
 - (c) Write a steps and syntax for following task in File: 06
 1. Adds a new item to the file.
 2. Modifies the details of an item.

OR

- Q - 5 Attempt following :** [14]
- (a) What is Derived attribute? Why are we considering them as derived? Write two examples of derived attributes. 06
 - (b) Differentiate the following: 04
 1. Activity and Operation
 2. Association and Aggregation
 - (c) How can encapsulation and polymorphism improve reusability? 04

- Q - 6 Attempt following :** [14]
- (a) What is significance of Activity diagram? Why we have to draw Activity diagram? Differentiate between in Activity diagram: 07
 1. Branch and Merge
 2. Fork and Join
 - (b) How Object diagram is differ from class diagram? Draw the Object diagram for ATM. 07

OR

- Q - 6 Attempt following :** [14]
- (a) What are the similarities/dissimilarities between a sequence diagram and collaboration diagram? Explain with suitable example covering all the aspects. 07
 - (b) What is significance of state chart diagram? Draw state chart diagram for ATM. 07
